

Multi-Agent System Design Document (MASD)

IntelliPlant AI

AI-Powered Industrial Knowledge Intelligence Platform

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Prepared By: Team IntelliPlant AI

Table of Contents

1. Introduction
 2. Objectives
 3. Multi-Agent Architecture
 4. Agent Hierarchy
 5. Agent Responsibilities
 6. Communication Protocol
 7. Agent Memory
 8. Tool Calling Architecture
 9. Agent Workflow
 10. Agent State Management
 11. Error Recovery
 12. Future Agent Expansion
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1. Introduction

IntelliPlant AI uses a Multi-Agent Architecture to solve complex industrial knowledge problems. Instead of relying on a single LLM, multiple specialized agents collaborate under a Supervisor Agent.

This approach improves:

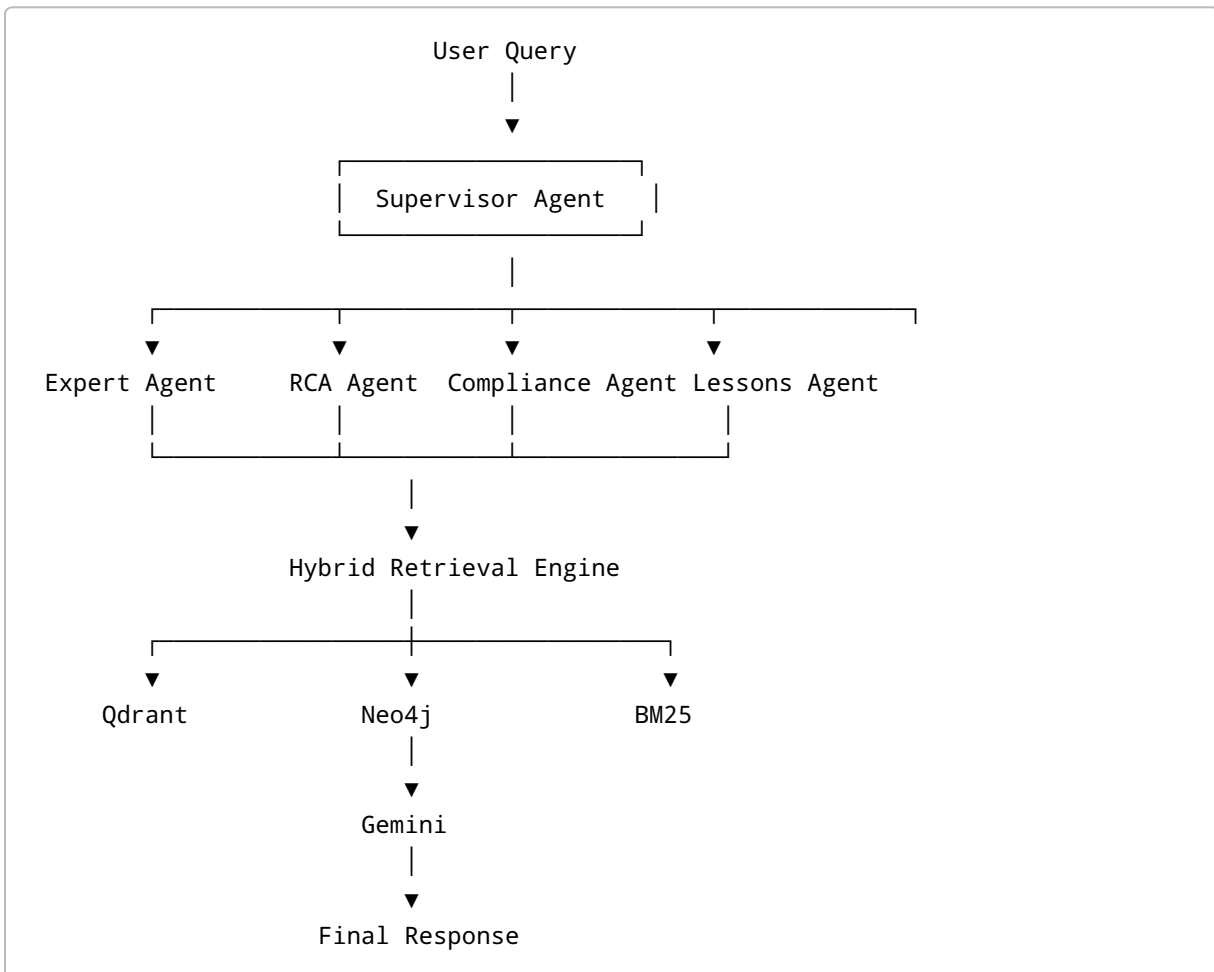
- Accuracy
 - Explainability
 - Reliability
 - Scalability
 - Fault tolerance
-

2. Objectives

The Multi-Agent System should:

- Decompose complex queries.
- Route tasks intelligently.
- Execute specialized reasoning.
- Prevent hallucinations.
- Support explainable responses.
- Enable future extensibility.

3. Overall Architecture



4. Agent Hierarchy

Level 1

Supervisor Agent

Level 2

Domain Agents

- Expert Copilot Agent
 - RCA Agent
 - Compliance Agent
 - Lessons Learned Agent
-

Level 3

Tool Agents

- Retrieval Agent
 - Graph Agent
 - Citation Agent
 - Memory Agent
-

Level 4

Infrastructure Agents

- Logging Agent
 - Monitoring Agent
 - Cache Agent
-

5. Supervisor Agent

Purpose

Acts as the brain of the system.

Responsibilities

Intent Detection

Determine query type.

Agent Selection

Choose specialized agents.

Context Management

Maintain conversation context.

Task Distribution

Parallel execution.

Response Aggregation

Combine outputs.

Functions

```
route_query()  
  
manage_context()  
  
merge_results()  
  
fallback_handler()
```

6. Expert Copilot Agent

Purpose

Engineering assistant.

Handles

- SOP retrieval
- Manuals
- Equipment information
- Operating procedures

Example Queries

"Show startup procedure for Pump P101."

"What is lubrication interval?"

"How to calibrate Valve V101?"

Tools

Vector Search

Graph Search

BM25 Search

Functions

```
retrieve_manual()
```

```
retrieve_sop()
```

```
generate_answer()
```

7. Root Cause Analysis Agent

Purpose

Failure analysis.

Responsibilities

Find probable causes.

Inputs

Incident reports

Maintenance history

OEM manuals

Inspection reports

Outputs

Root cause

Recommendations

Similar failures

Example Query

"Why Pump P101 failed?"

Functions

```
find_similar_incidents()
```

```
analyze_failure()
```

```
recommend_action()
```

8. Compliance Agent

Purpose

Regulatory intelligence.

Standards Supported

Factory Act

OISD

PESO

ISO 9001

ISO 45001

ISO 14001

Responsibilities

Compliance gap analysis

Audit evidence generation

Deviation detection

Functions

```
detect_gaps()
```

```
generate_compliance_report()
```

```
map_regulations()
```

9. Lessons Learned Agent

Purpose

Prevent repeated failures.

Inputs

Incident reports

Near misses

Audit findings

CAPA records

Responsibilities

Pattern analysis

Risk identification

Recommendation generation

Functions

```
analyze_patterns()
```

```
identify_repeated_failures()
```

```
generate_recommendations()
```

10. Retrieval Agent

Purpose

Unified search interface.

Search Sources

Vector Search

Qdrant

Graph Search

Neo4j

Keyword Search

BM25

Functions

```
vector_search()  
  
graph_search()  
  
keyword_search()  
  
context_fusion()
```

11. Citation Agent

Purpose

Explainability.

Responsibilities

Generate citations.

Track sources.

Assign confidence scores.

Output

```
{  
  "source": "manual.pdf",  
  "page": 15,  
  "confidence": 0.92  
}
```

12. Memory Agent

Purpose

Manage conversational memory.

Short-Term Memory

Redis

Stores

Recent interactions.

Long-Term Memory

Neo4j + PostgreSQL

Stores

Historical knowledge.

Semantic Memory

Qdrant

Stores

Embeddings.

Functions

```
retrieve_memory()  
store_memory()  
compress_context()
```

13. Logging Agent

Responsibilities

User logs

API logs

Agent logs

Error logs

Functions

```
log_request()  
log_response()  
track_errors()
```

14. Monitoring Agent

Purpose

Observability.

Metrics

Latency

Token usage

Agent performance

Failures

Cost

Functions

```
monitor_agents()

collect_metrics()

alert_failures()
```

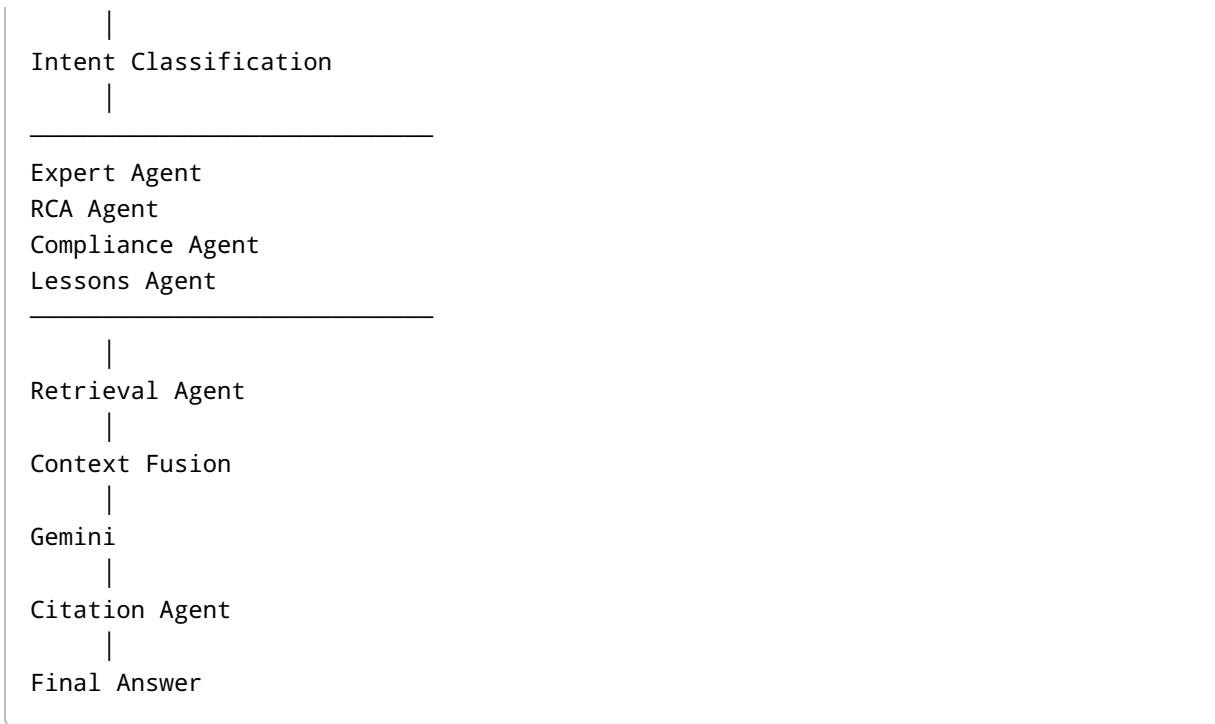
15. Communication Protocol

Agent interaction uses structured messages.

```
{
  "query": "Why Pump P101 failed?",
  "context": [],
  "intent": "RCA",
  "agent": "RCA Agent"
}
```

16. Agent Workflow

```
User Query
  |
  ▼
Supervisor Agent
```



17. Parallel Execution

Multiple agents may execute simultaneously.

Example:

Query

"Why Pump P101 failed and does it violate OISD guidelines?"

Parallel Agents

RCA Agent

+

Compliance Agent

Outputs merged by Supervisor Agent.

18. State Management

States

IDLE

RUNNING

WAITING

SUCCESS

FAILED

RETRYING

19. Error Handling

Agent timeout

↓

Supervisor retries.

LLM failure

↓

Fallback model.

Retrieval failure

↓

BM25 fallback.

Agent failure



Alternative agent selection.

20. Confidence Scoring

Final confidence

```
Confidence =  
0.4 Retrieval Score  
+  
0.3 Graph Evidence  
+  
0.3 Citation Score
```

Range

0-1

21. Future Agents

Predictive Maintenance Agent

Digital Twin Agent

Inventory Optimization Agent

SAP Agent

IoT Agent

SCADA Agent

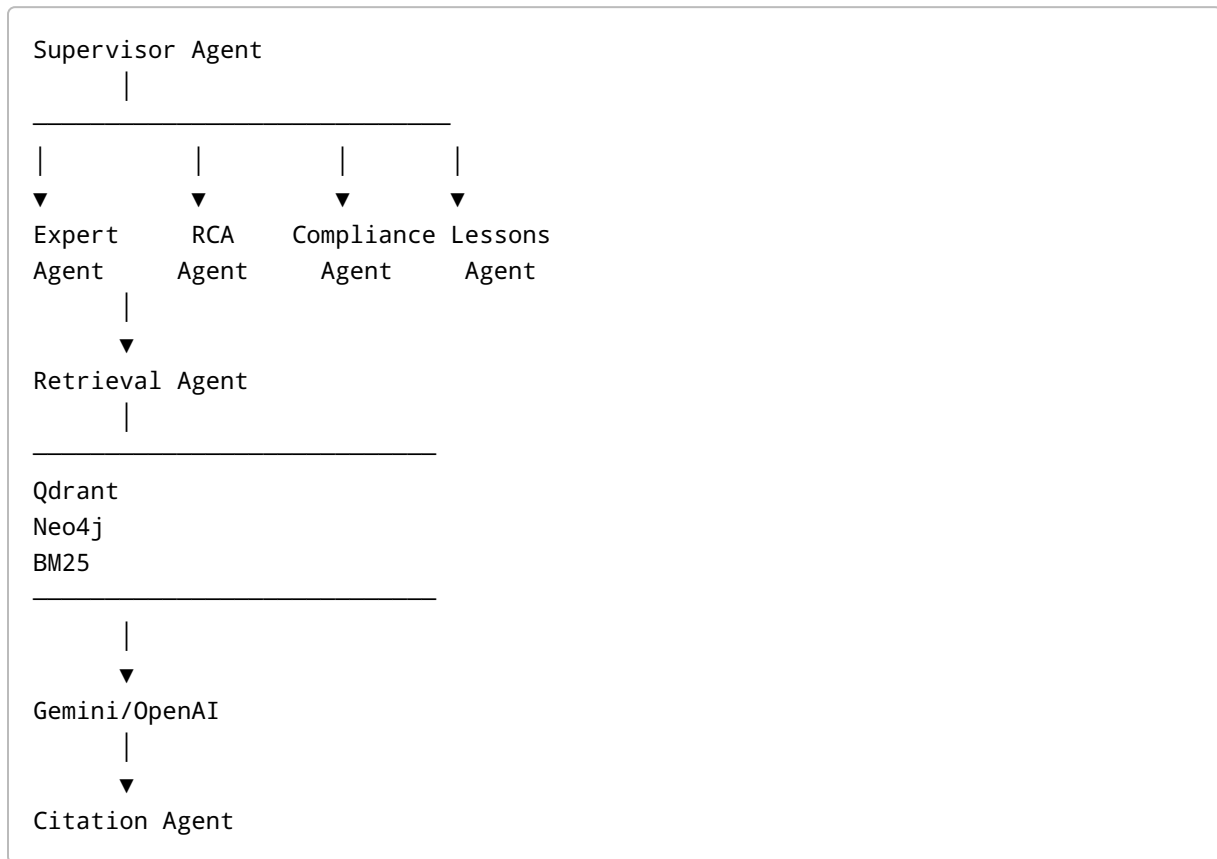
Cost Optimization Agent

Energy Efficiency Agent

Voice Copilot Agent

Autonomous Planner Agent

22. Agent Dependency Graph



23. Scalability

Horizontal agent scaling.

Independent agent deployment.

Asynchronous execution.

Distributed task queues.

Redis-backed memory.

24. Benefits

Higher accuracy.

Reduced hallucinations.

Explainable outputs.

Specialized reasoning.

Fault tolerance.

Future extensibility.

Parallel execution.

Conclusion

The Multi-Agent Architecture enables IntelliPlant AI to function as an Industrial Operations Brain by combining specialized agents, hybrid retrieval, memory systems, and explainable AI. The Supervisor Agent orchestrates domain-specific agents to deliver reliable, contextual, and citation-backed intelligence.